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Graham J. Sherwood

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Paleomagnetism and magnetostratigraphy of Miocene volcanics in eastern Otago and Banks Peninsula, New Zealand

GRAHAM J. SHERWOOD
Department of Geology
University College
P.O. Box 78
Cardiff, CF1 1XL, Wales*

Abstract Orientated samples were collected from 135 sites in 4 centres of Miocene alkali volcanic activity on the South Island of New Zealand: the Dunedin volcano, the Lyttelton volcano, the Akaroa volcano, and the Diamond Harbour Group. The natural remanent magnetisation (NRM) was measured using stepwise alternating field demagnetisation. From the results of at least three specimens a mean site NRM direction was calculated where possible. For each of the four centres a magnetostratigraphy has been determined. By comparing this with previously published K-Ar ages it has been possible to suggest correlations with the magnetic polarity timescale. The mean virtual geomagnetic poles (VGP) for the four centres, oldest to youngest, are: Dunedin volcano 86°.6N, 357°.0E; Lyttelton volcano 87°.7N, 086°.3E; Akaroa volcano 79°.7N, 137°.6E; and Diamond Harbour Group 68°.6N, 012°.1E. Even after rotation of the VGPs back to their original positions, using hotspot-relative rotations, the two youngest VGPs deviate significantly from the geocentric axial dipole position. The irregular nature of volcanic activity has apparently caused a bias towards a particular direction.

Keywords eastern Otago; Banks Peninsula; Miocene; volcanic rocks; paleomagnetism; magnetostratigraphy; natural remanent magnetisation; virtual geomagnetic poles; Dunedin volcano; Lyttelton volcano; Akaroa volcano; Diamond Harbour Group

*Present address: Institut für Allgemeine und Angewandte Geophysik der Universität München, Theresienstrasse 41, D-8000 München 2, West Germany.
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INTRODUCTION

On the eastern side of the South Island of New Zealand and the adjacent Chatham Rise and Campbell Plateau, there are several major centres of late Tertiary alkalic igneous volcanism, which show a general decrease in age towards the southeast (Adams 1981). As there appears to be no relationship between this activity and the plate boundaries, a linear mantle source has been proposed (Adams 1981; Wellman 1983). Four of these centres in South Island were selected for paleomagnetic study: the Dunedin volcano, the Lyttelton volcano, the Akaroa volcano, and the Diamond Harbour Group (Fig. 1). These all lie in regions which are considered not to have been affected by later tectonic activity (Walcott 1984). Three have already been the subject of paleomagnetic study: Dunedin (Coombs & Hatherton 1959), Akaroa (Evans 1970), and Stoddart Formation of the Diamond Harbour Group (Evans 1970).

All four centres have been dated by K-Ar (Stipp & McDougall 1968; Evans 1970; McDougall & Coombs 1973). The K-Ar ages quoted in this paper have been corrected from their original values for changes in the values of the decay constants (Steiger & Jager 1977), so are about 2.66% older than the published values. The errors quoted are given as two standard deviations and refer to experimental error rather than the reliability of the ages. K-Ar dates can be affected by a number of factors such as the enrichment or loss of radiogenic argon by alteration.

The Tertiary magnetic timescale is now well defined (e.g., Cox 1982) from studies of marine magnetic anomalies, paleomagnetism, and radiometric dates. The dates of the reversals have associated errors of about ±2%. By determining the magnetostratigraphy, it may be possible to get more information on the duration and timing of the igneous activity in these four volcanic centres.

Orientated cores were collected from 135 sites in the 4 areas. Four or five cores were taken from each site. They were orientated with the aid of a suncompass. A few were orientated by magnetic

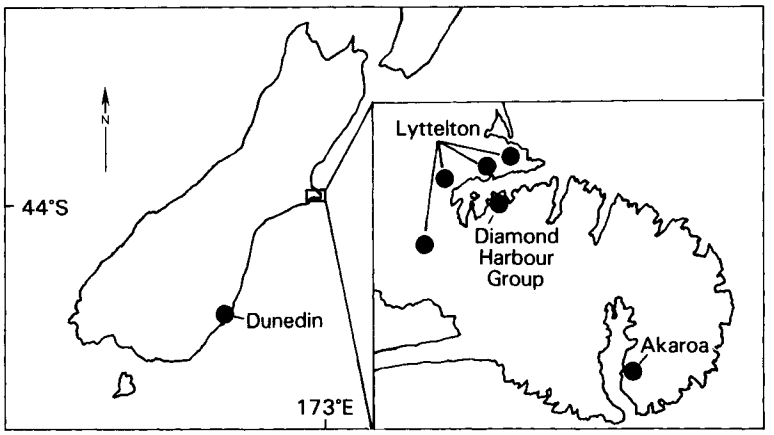


Fig. 1 South Island, New Zealand, showing the four areas of Miocene volcanic activity that were sampled.

means— in these, checks were made to correct for any anomalous magnetic effects caused by the rock body.

To discover how the ancient geomagnetic field direction changed during the periods of volcanic activity, the natural remanent magnetisation (NRM) of each flow was measured. In many flows the NRM was partially masked by a viscous remanent magnetisation (VRM) acquired recently; this was removed by low levels of alternating field (AF) demagnetisation.

At least three 25 mm diameter orientated samples from each flow were measured on a parastatic magnetometer. Samples were demagnetised in 5 mT steps up to 50 mT using a two axes tumbling alternating field system. The NRM directions were calculated using the LINEFIND computer programme (Kent et al. 1983). Site means were calculated using Fisher (1953) statistics.

Although rock magnetic analysis has indicated that in some flows the magnetic carriers have undergone low-temperature oxidation (see Sherwood 1988, this issue), for the purposes of magnetostratigraphy it will be assumed that the NRM direction represents the direction of the geomagnetic field at the time of eruption, or very near to it, unless there is good evidence to the contrary.

DUNEDIN VOLCANO

The alkali igneous rocks of the Dunedin volcano have been frequently studied by petrologists since the end of the last century. Marshall (1906, 1914) first described the sequences of basalts, phonolites, and trachytes that make up the igneous complex in the Dunedin area. Extensive research and mapping was carried out by Benson (e.g., 1941, 1942, 1968). A recent account of the geology is in Coombs et al. (1986).

The Dunedin volcano lies across the north end of the Taieri graben, which is bounded by high-angle northeast-trending reversed faults. These faults were probably active during the volcanic activity, since some outlying vents lie along the faults. Some of the faults were still active in the Pleistocene (Coombs 1965). Despite this faulting, the volcano remains largely undisturbed, with quaquaversal dips preserved in lava flows on all but the western flanks (Coombs 1983).

The complex shield which built up was about 30 km across, with many parasitic vents on its flanks. Benson (1942, 1968) divided the volcanism into four main phases. The Initial Eruptive Phase started with the eruption of basalt lavas (Allen 1974). These were followed by quartz-normative anorthoclase trachyte. In places, early basaltic and trachytic tuffs are interbedded with shallow marine sandstones and limestones of Middle Miocene age (Coombs et al. 1960). The rocks of the three Main Eruptive Phases (Benson 1968) are largely nepheline-normative alkali igneous rocks (Coombs & Wilkinson 1969). They range from olivine basalts through basanites and benmoreites to phonolites. The final stage of the volcanism was marked by the eruption of large phonolite domes, the remnants of which dominate the present landscape.

Radiometric dating has refined the Mid Miocene–Early Pliocene age determined by geological mapping (Coombs et al. 1960). Rb-Sr dates (Price & Compston 1973) give ages between

14.4 and 12 Ma, albeit with rather large errors. The K-Ar dating method has been used on nine rocks from the Dunedin igneous complex (McDougall & Coombs 1973). Dates of 13.4 ± 0.2 and 13.3 ± 0.4 Ma were obtained from the Initial Phase trachyte. No rocks were dated for the First Main Phase, but dates between 12.4 ± 0.4 and 11.6 ± 0.4 Ma came from rocks of the Second Main Phase. Basalts and phonolites from the Third Main Eruptive Phase gave dates between 11.6 ± 0.4 and 10.5 ± 0.2 Ma. So it appears that the volcano was active between 13.5 and 10.5 m.y. ago.

Rocks from the Dunedin volcano were extensively sampled during the late 1950s to determine their paleomagnetic directions (Coombs & Hatherton 1959; McDougall & Coombs 1973). The techniques used were relatively crude by modern standards in that in most samples were not magnetically cleaned. However, a reasonable number of samples was collected from each site. Rocks of the First Main Phase and the lower parts of the Second Main Phase (up to flow 21 (Marshall 1914) of the North Head sequence at Aramoana) were all of normal polarity. Those of the upper parts of the Second Main Phase had reversed or anomalous directions, whereas the Third Main Phase rocks were normally or anomalously magnetised. Coombs & Hatherton (1959) reported that many of the rocks, particularly the younger ones, did not have a stable NRM.

Thirty-one sites were sampled in the Dunedin area for the present study (Table 1, Fig. 2). These

Table 1 Paleomagnetic and sampling data for the Dunedin volcano.

Site code	Rock type*	Grid ref.	Mean declination	Mean inclination	Polarity	n/N	α_{95}	k
ZD1	3C	213866	NOT STABLE			0/4	–	
ZD2	2C	178756	NOT STABLE			0/5	–	
ZD3	2A	162814	78.7	78.2	I	3/5	6.5	360
ZD4	3C	187848	NOT STABLE			0/5	–	
ZD5	3B	136877	255.6	–76.2	I	4/5	16.0	34
ZD6	2B	136877	83.3	71.2	I	3/4	24.5	26
ZD7	2C	217775	306.9	–63.9	N	4/5	10.0	85
ZD8	2B	195779	355.8	–45.8	N	4/4	14.4	42
ZD9	3C	169872	NOT STABLE			0/5	–	
ZD10	3C	196855	NOT STABLE			0/6	–	
ZD11	2C	181792	30.2	–59.9	N	3/5	24.1	27
ZD12	0B	255867	NOT STABLE			0/5	–	
ZD13	3C	212882	NOT STABLE			0/5	–	
ZD14	2B	050746	NOT STABLE			0/5	–	
ZD15	2B	135878	38.3	53.3	I	4/5	22.4	18
ZD16	2B	134879	193.1	68.1	R	3/6	10.1	149
ZD17	2B	134880	32.3	–11.0	I	3/5	8.1	233
ZD18	2B	133880	203.7	58.0	R	3/5	13.0	91
ZD19	2B	133881	126.3	79.4	R	3/5	13.4	85
ZD20	2C	212919	NOT STABLE			0/5	–	
ZD21	2D	261891	165.0	69.3	R	3/5	17.3	52
ZD22	2B	333897	353.2	–66.1	N	3/5	2.1	3333
ZD23	1B	331888	53.4	–66.2	N	3/5	6.2	395
ZD24	1B	312869	344.9	–64.0	N	3/2	3.9	1000
ZD25	1D	284884	319.7	–72.5	N	3/3	21.3	34
ZD26	1B	311902	V. SCATTERED		N?	0/5	–	
ZD27	1B	308907	V. SCATTERED		N?	0/5	–	
ZD28	1B	308907	5.9	–63.7	N	2/2	3.4	5350
ZD29	1B	308907	5.6	–59.6	N	3/4	5.6	480
ZD30	2C	307910	4.5	–53.7	N	4/4	12.7	53
ZD31	2A	315915	V. SCATTERED		N?	0/4	–	

*A = trachyandesite, B = basalt, C = phonolite, D = benmoreite; 0, 1, 2, 3 refer to the Initial, First Main, Second Main, and Third Main Eruptive Phases (Benson 1968). Grid references refer to NZMS 260 series 1:50 000 topographic map sheets I44 and J44. n/N is the number of specimens used for the mean direction over the number of cores taken.

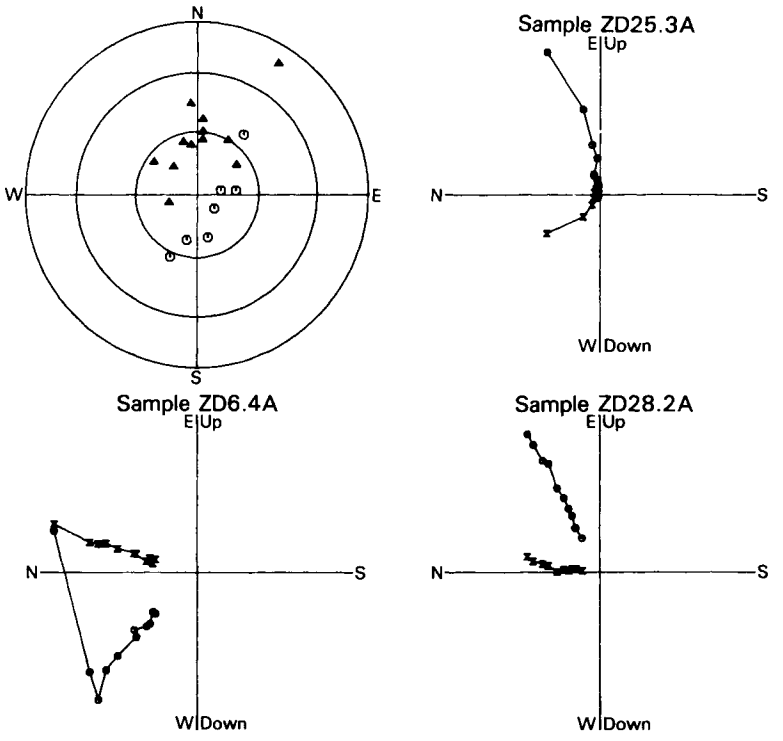


Fig. 2 Mean site NRM directions from the Dunedin volcano plotted on an equal area stereoplot: circles = positive inclination and triangles = negative inclinative. Orthogonal plots of some AF demagnetisation data (0 – 50 mT in 5 mT steps) are also shown: crosses represent the horizontal projection, and circles the vertical projection.

were a representative sample of the entire igneous sequence. Twelve of the 31 sites failed to provide a stable NRM direction. This included all five of the Third Phase phonolites and also the Initial Phase basalt at Careys Bay. Two of the First Phase basalts from Aramoana (ZD26, ZD27) had NRMs that were generally of normal polarity; some cores had apparently stable NRMs with completely different (including reversed) directions, probably because of late low-temperature oxidation in some parts of the flows (Sherwood 1988, this issue).

It has not proved possible to ascertain the polarity during the Initial Eruptive Phase. The First Main Phase and the lower part of the Second Phase are of normal polarity (sites ZD7, 8, 11, 22, 23, 24, 25, 28, 29 and 30). From the middle of the Second Phase to its top, the sequence seems to be of reversed polarity. The Leith Valley Andesite (Benson 1968) (site ZD3) has an intermediate direction with a reversed inclination. (NRM directions whose virtual geomagnetic poles (VGPs) are >40° from the Earth's rotation axis are considered to be intermediate.) Coombs & Hatherton (1959) reported that the underlying Roslyn Dolerite has a reversed NRM, suggesting that the Leith Valley Andesite may have recorded a magnetic excursion.

Three of the seven flows of the Swampy Summit sequence (Fig. 3) have a reversed NRM, the others have intermediate NRMs. The NRM directions determined are in good agreement with those reported in McDougall & Coombs (1973). Although this part of the volcano has been affected by faulting, the effect on the NRM directions is not obvious. The bottom two flows (ZD19, 18) have reversed NRMs. The next flow (ZD17) has an intermediate direction with a shallow normal inclination. This is followed by a flow with a reversed NRM (ZD16), on top of which lie two flows (ZD15, 6) with similar intermediate NRMs (with reversed inclinations and easterly declinations). The top flow (ZD5) has a westerly declination and normal inclination.

The boundary between the Second and Third Main Phases lies between the top two flows of the sequence (McDougall & Coombs 1973) and is marked by a R-N transition. The third flow from the bottom of the section (ZD17) records an earlier magnetic excursion (or perhaps a reversal). Benson (1968) mapped all the lower flows of the Swampy Summit section as belonging to the upper part of the Second Main Eruptive Phase, which is supported by the reversed polarity of the bottom flows. The K-Ar date from the bottom flow (12.4 ± 0.4 Ma) is,

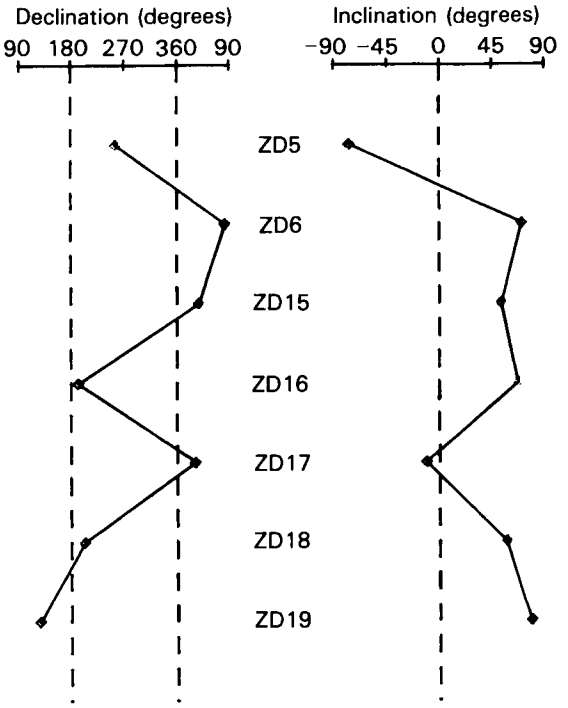


Fig. 3 Stratigraphic variation of NRM direction in the Swampy Summit section (vertical scale is arbitrary).

however, some 0.8 Ma older than the dates from the top two flows (McDougall & Coombs 1973). The date from this bottom flow is probably incorrect, because of excess radiometric argon.

No Third Phase flow samples, apart from the top flow of the Swampy Summit section, had a stable NRM, but from the R-N reversal it appears likely that at least the bottom part of the Third Phase is of normal polarity.

The magnetostratigraphy seems to be normal-reversed-normal, with the reversals occurring in the middle and the end of the Second Main Phase. This agrees with the findings of Coombs & Hatherton (1959). There are signs (sites ZD3, 17) of at least one R-N-R excursion during the upper part of the Second Main Phase.

The K-Ar dates (McDougall & Coombs 1973) from the Dunedin volcano all lie between 13.4 and 10.5 Ma. The predominant polarity in that time interval was reversed (Cox 1982). The problem is to reconcile the observed magnetostratigraphy with the K-Ar dates and the polarity timescale (Fig. 4). Nothing can be said about the Initial Eruptive Phase as there are no paleomagnetic data prior to the First Main Phase. It seems likely that the First Phase and lower parts of the Second Main Phase erupted during the same interval of normal polarity. The only K-Ar dates available from this period are from a Second Phase flow at Aramoana (site ZD30). Two determinations from this sample gave ages of 12.2 ± 0.4 and 11.9 ± 0.4 Ma (McDougall & Coombs 1973). This makes correlation with chron 5A.2 likely (12.03–11.77 Ma, Cox 1982). Dates for normally magnetised samples from the Third Main Phase are 11.3 ± 0.4 and 11.5 ± 0.4 Ma (McDougall & Coombs 1973). The normal chron which best fits these dates is 5A.1 (11.63 – 11.47 Ma, Cox 1982).

The correlations suggest that the main part of the activity of the Dunedin volcano occurred between 12 and 11.5 Ma, but more radiometric dates should be obtained to verify this.

BANKS PENINSULA

Banks Peninsula (inset Fig. 1), which lies to the southeast of the city of Christchurch, is made up of the remains of two large shield volcanoes: Lyttelton, the older, forms the northwestern part of the peninsula, and Akaroa forms the eastern part. As well as these two volcanoes, there was smaller scale intermittent volcanicity mainly in the northern part of the peninsula around the settlement of Diamond Harbour. From K-Ar dating, the igneous activity lasted 6 m.y. (Stipp & McDougall 1968), starting c. 12 m.y. ago.

The geology and petrology of the peninsula was studied in detail by Speight (e.g., 1917, 1924, 1938, 1944). The stratigraphy of the late-stage activity was revised by Liggett & Gregg (1965). Recent detailed mapping of the central part of the peninsula has led to further changes to the stratigraphy (Sewell 1988). Geochemical and petrological studies by Weaver & Sewell (1986) have shown that the igneous activity may be classed as transitional to mildly alkalic.

LYTTELTON VOLCANO

The cone of the Lyttelton volcano is deeply dissected by a northeast-trending elongate erosion crater, Lyttelton Harbour. On the southwestern side of the volcano, in the Gebbies Pass area, the underlying rocks are exposed. The basement is composed of greywackes of the Torlesse Group (here Late Paleozoic). The earliest volcanic eruptions were of the McQueens Andesite and associated rhyolites (now known from Rb-Sr data

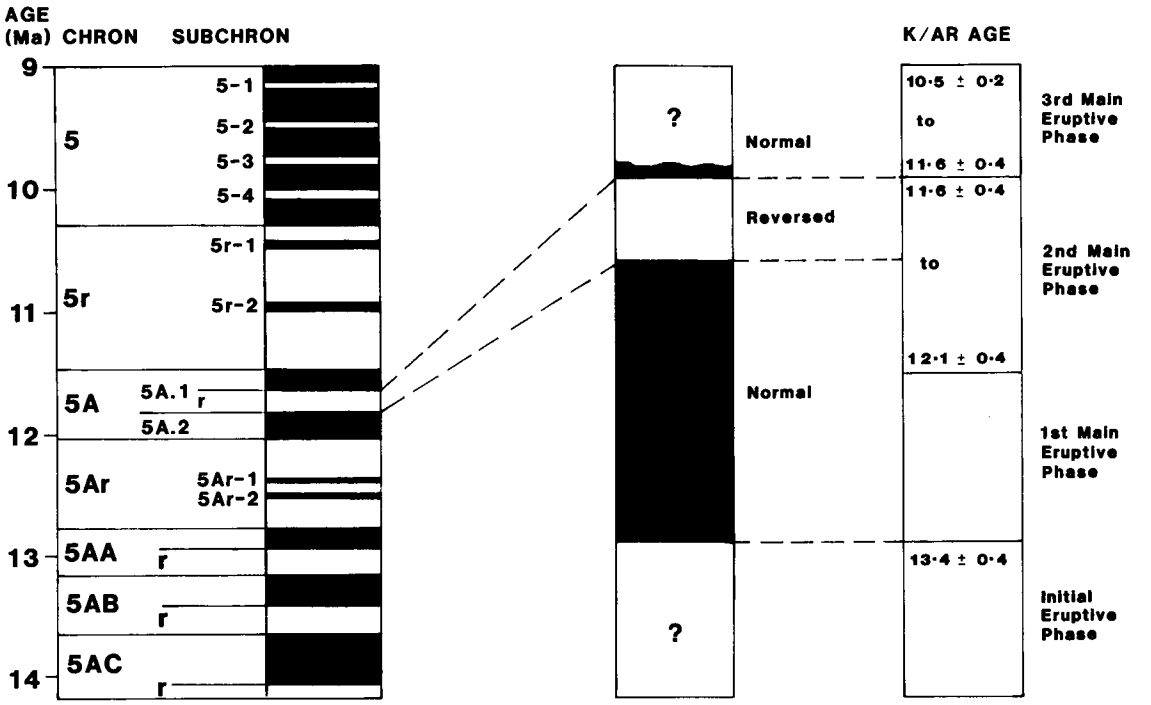


Fig. 4 The magnetostratigraphy of the Dunedin volcano and its suggested correlation with the magnetic timescale (Cox 1972). K-Ar dates are recalculated from McDougall & Coombs (1973).

to be Late Cretaceous—Weaver & Sewell 1986); these are covered by the Paleocene Charteris Bay Sandstone. Rhyolites and andesites of the Governors Bay Volcanics (Weaver & Sewell 1986) overlie the sandstone: these immediately predate the formation of the Lyttelton volcano. The rhyolites have yielded a Rb-Sr date of 10.8 ± 0.1 Ma (Barley et al. cf. Weaver & Sewell 1986).

The Main Dome of Lyttelton volcano is composed largely of basic lavas with thin pyroclastic layers. The lavas are mainly plagioclase-phyric hawaiites with a few true basalt, mugearite, and trachyte flows (Weaver 1980). K-Ar dates from the Main Dome Phase range from 12.2 ± 0.4 to 10.6 ± 0.3 Ma (Stipp & McDougall 1968).

There was a late period of activity near the Tors on the Summit Road, where lavas and lahars lie in a channel cut into the Main Dome lavas (Weaver 1980). These late lavas are cut by trachyte dikes.

Dikes ranging from basalt to trachyte in composition were emplaced throughout the history of the volcano, and today they account for about 2.5% of the rocks along the Summit Road (Speight 1938). A late-stage radial dike swarm, centred in Charteris Bay (Frost 1965), accounts for a majority of the trachyte dikes. One of these was dated by Stipp & McDougall (1968) at 9.93 ± 0.29 Ma.

Samples were collected from 30 sites on the western side of the volcano, mainly along the Summit Road. Three were from late-stage trachyte dikes, two from the well-known multiple dikes near the Sign of the Kiwi (Speight 1938; Liggett & Gregg 1965; Weaver & Sewell 1986), and the rest from lava flows with a range of stratigraphic positions (Table 2, Fig. 5). Only 2 out of the 30 sites did not have a stable NRM. Most of the lavas had NRMs of reversed polarity, the exceptions being the two sites ZBL29 and 30, which are very high in the stratigraphic sequence, and may postdate the Main Dome Phase. Flow ZBL2, which lies at the very bottom of the stratigraphic sequence, has recorded an intermediate/reversed direction. The three trachyte dikes, ZBL7, 15, and 22, were all of normal polarity, but the two parts of the multiple dikes had very different NRM directions.

The main trachyte dike (ZBL24) gave an anomalous intermediate direction with a normal inclination and a southerly declination, and the

mugearite dike (ZBL25) had a reversed NRM. Thermomagnetic analysis (Sherwood 1988, this issue) suggests that the magnetic carrier in ZBL24 is magnetite and in ZBL25 is cation-deficient titanomagnetite. This suggests that the NRM in the large dike (ZBL24) is primary, but the fact that the dike has undergone severe alteration, probably geothermal (Weaver pers. comm.), makes a chemical origin for the magnetite and the anomalous direction more likely. The trachyte dike (ZBL24) has been correlated with a lava in Dyers Pass, and it is believed that the dike fed the flow (Weaver pers. comm.). This means that this dike belongs to the Main Dome Phase. There is no good evidence as to the age of the mugearite dike (ZBL25): both it and the trachyte dike have chilled margins, and they are separated by a sliver of country rock (hawaiite). It is thought likely that it belongs to the Main Dome Phase (Weaver pers. comm.).

Lavas of the Main Dome Phase seem to have erupted during a period of reversed polarity. The topmost lavas formed during a period of normal polarity, as did the late-stage trachyte dikes.

A reversed polarity period followed by a period of normal polarity is the minimum magnetostratigraphy required for the Lyttelton volcano. If the mugearite dike was younger than the late lavas then it would be necessary to insert an additional period of reversed polarity. This model is less likely than the simple R-N sequence. In the preferred magnetostratigraphy (Fig. 6) there is a long period of predominantly reversed polarity (chron 5r) from 11.47 to 10.3 Ma (according to Cox 1982), with two normal subchrons, the 5r-2 (10.91–10.99 Ma) and 5r-1 (10.43–10.48 Ma). This is followed by a period of mainly normal polarity (chron 5) from 10.3 to 8.98 Ma, with four short reversed subchrons (5-1 to 5-4).

Table 2 Paleomagnetic and sampling data for the Lyttelton volcano.

Site code	Stratigraphic position	Grid ref.	Mean declination	Mean inclination	Polarity	n/N	α_{95}	k
ZBL1	M	089532	166.6	56.7	R	3/8	7.5	270
ZBL2	M	992382	120.9	89.5	I	3/5	4.8	660
ZBL3	M	992382	150.5	79.7	R	3/3	5.1	587
ZBL4	M	992385	145.5	62.0	R	3/5	6.0	420
ZBL5	M	988433	143.2	70.7	R	3/5	6.0	418
ZBL6	M	073492	180.8	32.8	R	3/3	6.3	385
ZBL7	D	074492	346.3	–65.2	N	3/3	3.5	1250
ZBL8	M	074492	185.2	58.1	R	3/5	2.5	2400
ZBL9	M	074492	174.7	46.1	R	4/5	19.2	24
ZBL10	M	075492	185.4	56.3	R	3/5	10.7	133
ZBL11	M	075092	173.1	41.6	R	3/5	34.1	14
ZBL12	M	076092	187.1	53.1	R	3/3	6.7	340
ZBL13	M	076492	203.6	72.0	R	4/5	10.0	86
ZBL14	M	077492	197.2	64.4	R	3/5	7.3	285
ZBL15	D	072491	359.3	–66.8	N	4/5	21.0	20
ZBL16	M	992445	140.9	64.7	R	3/3	8.1	233
ZBL17	M	112507	148.1	40.2	R	3/5	12.3	101
ZBL18	M	102483	145.1	66.7	R	4/5	22.8	17
ZBL19	M	102482	182.7	53.4	R	3/5	6.1	410
ZBL20	M	098481	192.4	62.9	R	3/6	7.9	244
ZBL21	M	097484	188.9	64.7	R	4/4	11.4	66
ZBL22	D	098487	18.6	–71.2	N	3/5	8.2	277
ZBL23	M	088493	159.9	53.8	R	3/5	8.9	192
ZBL24	D	008465	196.0	–73.3	I	4/5	13.7	46
ZBL25	D	008465	176.9	66.0	R	3/2	9.2	179
ZBL26	M	006494	NOT STABLE			0/5	–	
ZBL27	M	005464	196.4	53.0	R	3/4	8.3	222
ZBL28	M	012477	NOT STABLE			0/5	–	
ZBL29	L	025482	18.4	–52.6	N	3/4	11.4	118
ZBL30	L	023476	41.0	–66.4	N	3/4	9.6	166

*M = Main lava, L = late lava, D = dike; grid references refer to NZMS 1 series topographic map sheet S84.

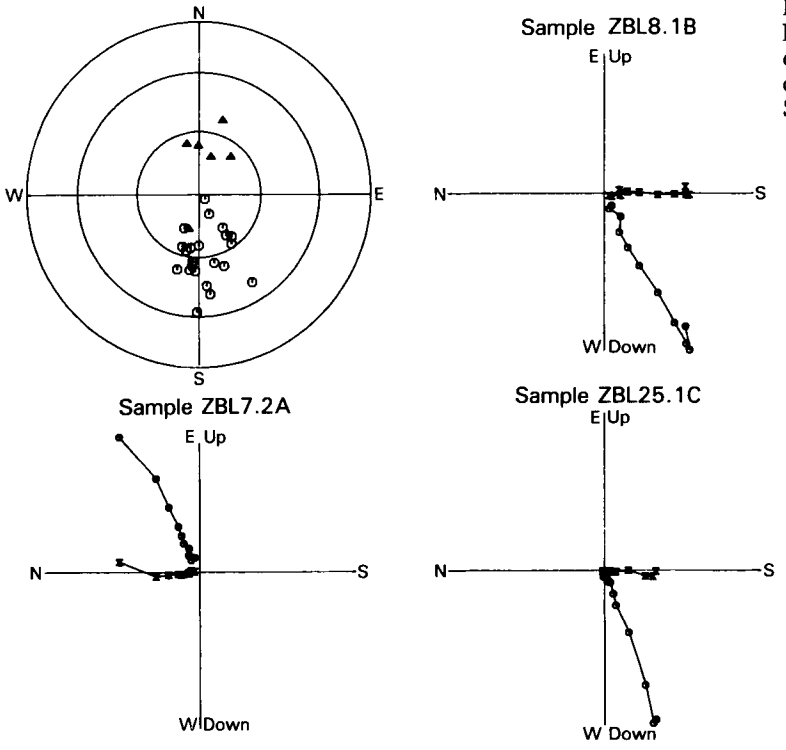


Fig. 5 Stereoplot of mean site NRMs for the Lyttelton volcano and examples of AF demagnetisation data (0–50 mT in 5 mT steps). Symbols as in Fig. 2.

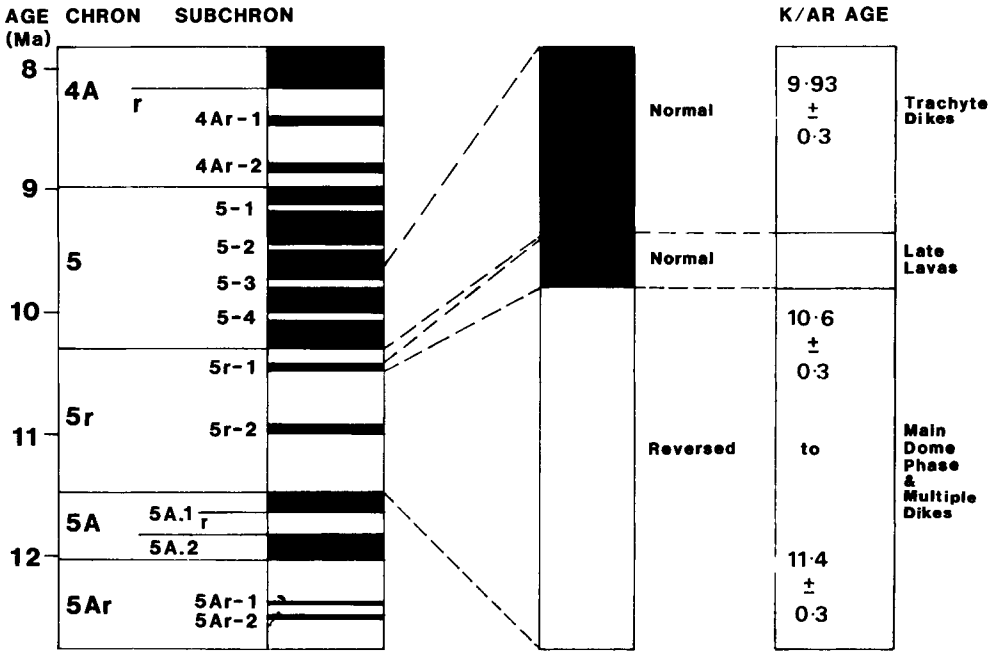


Fig. 6 A possible correlation of the magnetostratigraphic sequence for the Lyttelton volcano with the magnetic timescale (Cox 1982). K-Ar dates are recalculated from Stipp & McDougall (1968).

The Main Dome Phase of the Lyttelton volcano probably occurred during magnetochron 5r. Given the random sampling of lavas, and the irregular nature of volcanic activity, it is possible that short periods of opposite polarity have not been picked up in this study. The Rb-Sr date of 10.8 ± 0.1 Ma from the underlying rhyolites is thought to be a crystallisation age (Weaver pers. comm.). If so, this would throw into doubt most of the K-Ar ages of Stipp & McDougall (1968). However, the age from the bottom of the Main Dome Phase near Gebbies Pass does seem to be anomalously old at 12.3 ± 0.4 Ma when compared with the magnetic timescale. It is likely that this K-Ar date is incorrect. With the intermediate direction of ZBL2 it seems likely that the first lavas erupted immediately after a N-R transition. With the available radiometric dates it is possible to correlate the base of the Lyttelton Main Dome Phase with the bottom of chron 5r (11.47 Ma according to Cox 1982). The late-stage lavas may be recording the normal subchron 5r-1 (10.48 –10.43 Ma) or possibly the normal polarity towards the bottom of chron 5 (after 10.3 Ma). The late-stage trachyte dikes have a single K-Ar date of 9.93 ± 0.29 Ma which puts them in the oldest third of chron 5.

AKAROA VOLCANO

The lavas of the Akaroa volcano are mildly alkalic basalts and hawaiites (Price & Taylor 1980). A dike swarm, similar to that at Lyttelton, cuts through the lavas, but is less well exposed. There are also a number of late trachyte domes intruded through the lava pile (Speight 1944). At the very heart of the volcano, on Onawe Peninsula, gabbro and syenite, the only plutonic rocks seen on Banks Peninsula, are exposed (Weaver & Sewell 1986).

Stipp & McDougall (1968) dated a wide range of lavas and the Onawe intrusives by the K-Ar method. Evans (1970) also used K-Ar to date samples from the lava sequence on the Lighthouse Road above Akaroa. Over 30 samples from the Akaroa volcano have been dated. Apart from a date of 12.1 Ma from the syenite on Onawe, which was thought to be anomalously old, and an anomalously young age for one site on the Lighthouse Road, the intrusives and lavas gave dates ranging from 9.4 to 8.2 Ma.

The Lighthouse Road, to the south of Akaroa village, has been extensively sampled by paleomagnetists. The 685 m succession of probably over 100 flows was first sampled by Evans (1970), who collected from 73 flows from the entire section, starting at Green Point on the shore of Akaroa Harbour. He determined a N-R-N-R-N-R magnetostratigraphic sequence. Hoffman (1986) resampled parts of the section, in particular the N-R and R-N transitions in the middle of the sequence (at around 410 m and 570 m, respectively). His measurements indicate that there may have been up to three excursions of the magnetic field between these two reversals. Intermediate directions for both these transitions have been recorded in the lava sequence, indicating a reasonably fast extrusion rate for lavas at those times.

Samples were taken from 57 flows on the Lighthouse Road section between 340 and 670 m above sea level (Table 3, Fig. 7). Only flows which were fresh and clearly *in situ* were sampled. Exposure was better in the top part of the section — only nine flows were sampled in the bottom 160 m. All flows dip at about 10° to the south, away from the centre of the volcano. The flows sampled were all basalt and hawaiite aa flows, with rubbly tops and bottoms and massive centres.

Only one flow (ZBA51) did not have a stable NRM. The top flow (ZBA1) had some cores with a very large secondary component superimposed upon the NRM. This is interpreted as being caused by a lightning strike. It was quite easily removed by AF demagnetisation to leave a reversed NRM. Samples from flow ZBA38 also have a strong secondary component; when this is removed the NRM component remaining has a slightly anomalous direction. It is not clear whether a genuine intermediate direction has been recorded (possibly after eruption as maghemitisation has occurred), or whether part of the secondary magnetisation has not been removed.

The magnetostratigraphic succession (Fig. 8) is broad agreement with those of Evans (1970). There are three transitions recorded in this section: N-R, R-N and N-R. The bottom two transitions have been described by Hoffman (1986). The sampling of the bottom transition was incomplete in the present study, with no samples taken between 356 and 420 m, because of poor exposure. The NRM directions from the second transition (Fig. 9) show similar behaviour to that found by Hoffman (1986). A shallow reversed direction is recorded in several flows prior to the actual reversal. The top reversal has no intermediate directions recorded between the perfectly normal (ZBA17), and the perfectly reversed flow which lies directly above (ZBA16).

Evans (1970) correlated the 410 m N-R transition with the top of chron 5 (8.98 Ma, Cox

Table 3 Paleomagnetic and sampling data for the Akaroa volcano.

Site code	Altitude (m a.s.l.)	Grid ref.	Mean declination	Mean inclination	Polarity	n/N	α_{95}	k
ZBA1	670	294167	166.3	65.2	R	3/5	10.0	152
ZBA2	660	294167	192.5	60.0	R	4/4	14.7	40
ZBA3	644	293169	192.2	47.8	R	3/5	10.4	141
ZBA4	643	293169	175.3	48.0	R	4/7	10.6	76
ZBA5	642	293169	171.2	41.2	R	4/6	9.7	90
ZBA6	641	293169	167.3	46.6	R	3/6	6.9	320
ZBA7	640	293169	151.4	46.4	R	3/5	7.0	311
ZBA8	635	293170	136.7	49.4	R	3/5	5.9	435
ZBA9	632	293170	148.6	38.9	R	3/5	5.4	520
ZBA10	630	293170	144.6	36.8	R	4/5	21.1	20
ZBA11	628	293171	173.4	58.4	R	3/4	8.5	211
ZBA12	623	293171	164.0	58.8	R	4/6	15.3	37
ZBA13	620	293171	171.3	59.5	R	4/5	7.7	143
ZBA14	616	293172	192.4	61.3	R	3/5	7.8	250
ZBA15	613	293172	163.0	49.7	R	4/4	9.0	105
ZBA16	611	293172	167.8	57.8	R	3/5	5.7	470
ZBA17	609	293172	18.8	–67.2	N	4/5	13.9	45
ZBA18	607	293172	15.9	–49.0	N	3/5	9.1	183
ZBA19	600	293172	13.5	–49.6	N	3/6	8.1	233
ZBA20	597	293173	19.2	–46.6	N	4/6	13.3	49
ZBA21	595	293173	6.3	–53.5	N	3/5	6.5	360
ZBA22	592	293174	3.7	–77.8	N	3/5	10.2	146
ZBA23	590	293174	340.1	–71.7	N	3/6	5.7	460
ZBA24	588	293174	339.1	–38.8	N	3/6	3.3	1390
ZBA25	584	294175	341.3	–47.2	N	3/5	12.6	97
ZBA26	580	294177	354.5	–47.7	N	3/5	8.7	201
ZBA27	578	294178	285.2	–78.6	I	3/4	8.6	206
ZBA28	574	294179	6.3	–60.5	N	3/5	11.3	119
ZBA29	572	294179	151.8	–13.4	I	3/5	9.2	180
ZBA30	570	293180	185.0	31.6	R	4/5	12.3	57
ZBA31	567	293180	176.9	30.4	R	3/8	15.5	64
ZBA32	564	293180	175.0	22.0	R	4/6	14.5	41
ZBA33	561	293180	176.7	18.2	R	3/6	8.1	232
ZBA34	560	293181	168.4	31.8	R	4/3	9.7	91
ZBA35	557	293181	175.0	49.0	R	3/4	9.8	160
ZBA36	553	293181	165.2	64.1	R	3/5	7.6	263
ZBA37	547	293182	168.0	29.9	R	4/5	25.6	14
ZBA38	545	293182	239.5	59.1	I	4/3	16.7	31
ZBA39	542	293182	159.5	47.7	R	3/3	4.7	680
ZBA40	539	294182	145.0	58.4	R	4/5	17.5	28
ZBA41	536	294182	149.7	64.7	R	3/5	16.9	54
ZBA42	532	294183	177.5	64.7	R	3/5	7.2	295
ZBA43	529	294183	171.1	55.6	R	4/5	16.4	32
ZBA44	525	295184	143.4	61.6	R	2/1	3.1	6411
ZBA45	515	295185	144.6	57.4	R	3/8	12.2	102
ZBA46	510	295185	187.6	53.1	R	4/4	17.8	28
ZBA47	505	295185	174.0	59.2	R	3/7	4.9	630
ZBA48	500	294187	202.0	54.9	R	4/2	11.2	68
ZBA49	495	292190	185.3	76.5	R	3/5	11.8	110
ZBA50	485	292191	135.6	60.4	R	3/5	14.9	69
ZBA51	460	292192	NOT STABLE			0/7	–	
ZBA52	450	292192	194.8	39.7	R	3/5	9.9	156
ZBA53	430	292192	176.5	43.2	R	3/4	12.5	98
ZBA54	428	292192	171.5	49.7	R	3/5	8.3	222
ZBA55	420	291193	198.7	61.2	R	3/6	16.2	59
ZBA56	356	288196	27.0	–61.1	N	4/4	15.7	35
ZBA57	340	286198	54.6	–18.1	I	3/6	9.0	188

Grid references refer to NZMS 1 series topographic map sheet S94. Altitudes are from NZMS 270 series 1:25 000 topographic map sheet N37A.

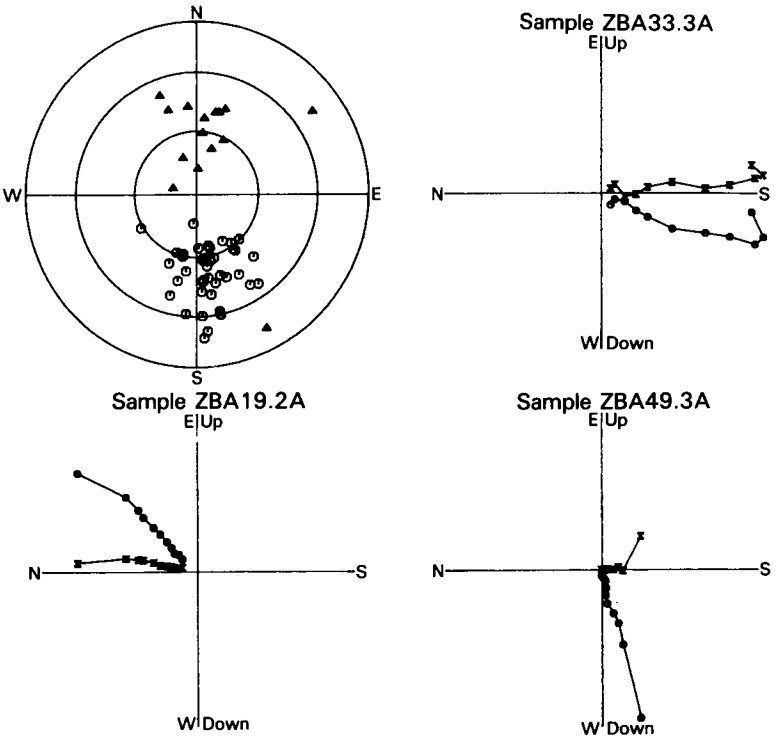


Fig. 7 Stereoplot of mean site NRMs from the Akaroa volcano, along with examples of AF demagnetisation behaviour (0 – 50 mT in 5 mT steps). Symbols as in Fig. 2.

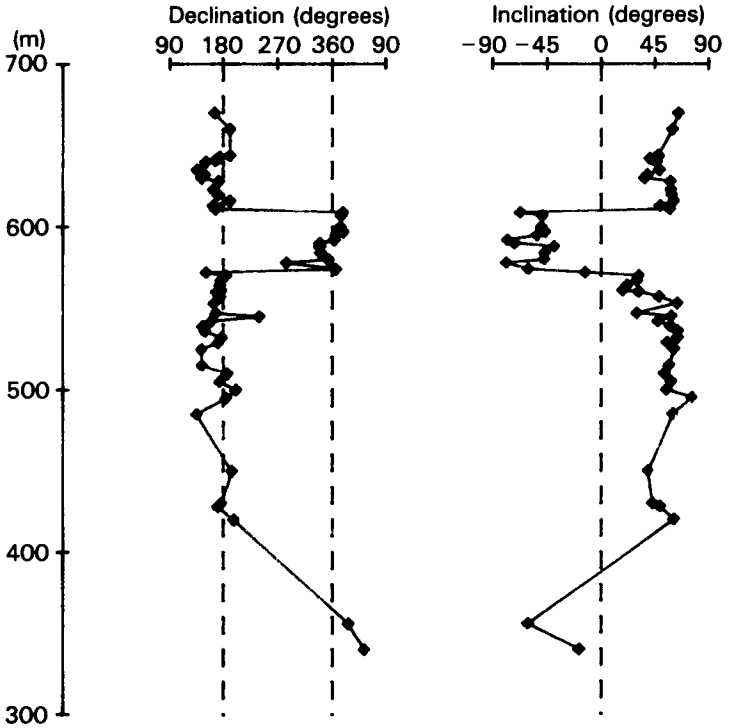


Fig. 8 The stratigraphic variation of mean flow NRM directions in the Lighthouse Road section (vertical scale height above sea level).

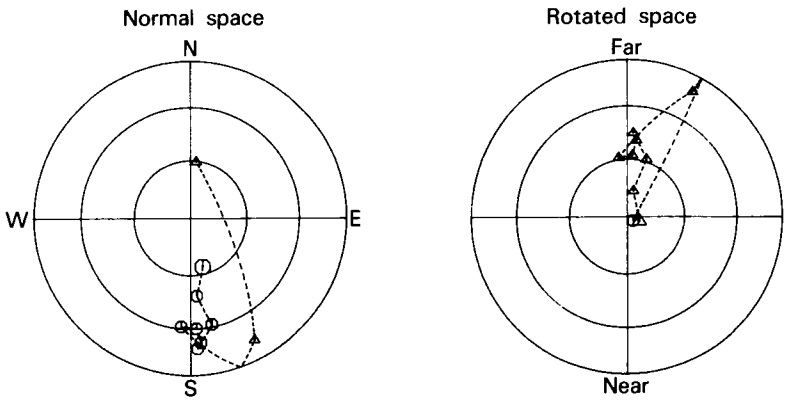


Fig. 9 An R-N transition has been recorded in detail in the Akaroa section: a reversed direction with a shallow inclination is recorded in several flows prior to the actual reversal. The transition is plotted on Schmidt stereonets both in normal space and in D-, I- space (Hoffman 1984) centred on the expected geocentric axial dipole position for Akaroa. Symbols as in Fig. 2.

1982), and the topmost normal interval with the normal subchron 4Ar-2 (8.80–8.87 Ma). Hoffman (pers. comm.) has suggested that this top R-N transition corresponds to the the base of chron 4A at 8.18 Ma (Cox 1982). Another possibility is that the top normal period corresponds to the normal subchron 4Ar-1 (8.40 – 8.49 Ma, Cox 1982), with the lower N-R transition of the section (this study) marking the top of subchron 4Ar-2.

From the mean K-Ar dates calculated for the section (Evans 1970) the mean age for the bottom of the section (this study) is 9.0 ± 0.2 Ma and for the top is 8.6 ± 0.2 Ma. This makes correlation of the top normal period with subchron 4Ar-2 (8.80 – 8.87 Ma) the most likely (Fig. 10). This was the correlation favoured by Evans (1970). The lower N-R transition will correlate with the top of chron 5 and the reversed interval found by Evans (1970) at about 70 m above sea level will correspond to subchron 5-1.

DIAMOND HARBOUR GROUP

The last stages of volcanism on Banks Peninsula were fairly localised eruptions (Sewell 1988), often infilling valleys eroded in the earlier Lyttelton flows. The activity was mainly centred around Quail Island and the Diamond Harbour area.

Liggett & Gregg (1965) divided these late alkali basalts into three formations which together comprised their Diamond Harbour Group. The Church Formation usually has well-developed columnar jointing in its flows, whereas the Kaioruru Formation appears coarse grained, often forming pahoehoe flows, and the Stoddart Formation forms aa flows with well-developed flow breccias. The three formations are on the shore at Church Bay, with angular unconformities between them.

The erosional contacts indicate that the three formations were formed at separate times and volcanism was not continuous. The Church Formation has a fluvial conglomerate unit between two of its flows at Church Bay. K-Ar dating (Stipp & McDougall 1968; Evans 1970) supports the episodic nature of the volcanism: the Church flow at Purau Bay gives an age of 8.24 ± 0.2 Ma; a Kaioruru flow at Church Bay is dated at 7.03 ± 0.2 Ma; and flows from the largest sequence of Stoddart flows which forms the dip-slope from Mount Herbert to Diamond Harbour have all given ages of c. 6 Ma, with no difference between the flows at the top and the bottom (Evans 1970), even though this part of the Stoddart Formation is over 300 m thick with in excess of 50 flows.

Evans (1970) sampled a sequence of 53 flows from this part of the Stoddart Formation. He found that the whole sequence had a similar slightly anomalous, almost transitional, reversed direction. From this he concluded that eruption must have been extremely rapid, which is supported by the K-Ar data.

Recent mapping of the Mount Herbert and Diamond Harbour area has led to revision of the stratigraphy (Weaver & Sewell 1986; Sewell 1988). The Church Formation is no longer considered to be part of the Diamond Harbour Group (Sewell 1988), but for the purposes of this paper results from the Church Formation will be included with the Diamond Harbour Group.

Samples were taken from two flows of the Church Formation: one at Purau Bay and one at Church Bay. Three Kaioruru flows were also sampled at Church Bay. Twelve flows were sampled from the Stoddart Formation, from various localities in Diamond Harbour and Church Bay (Table 4, Fig. 11).

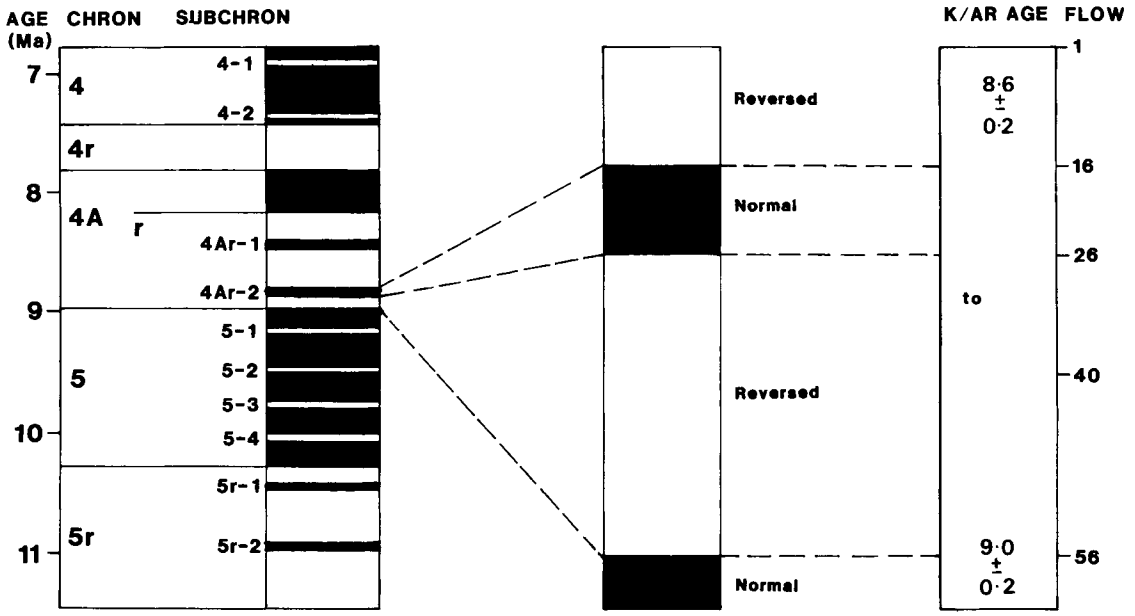


Fig. 10 Proposed correlation of the magnetostratigraphy of the Lighthouse Road section through the Akaroa volcano with the magnetic timescale (Cox 1982). The K-Ar ages are the mean ages for the top and bottom of the section (recalculated from Evans 1970).

The Church flows both have normal NRMs. Two of the Kaioruru flows are of reversed polarity, and the third records an intermediate direction. The Stoddart flows all have directions that are reversed or borderline between reversed and intermediate.

Assuming the K-Ar dates are approximately correct, it is possible to make tentative correlations with the reversal timescale (Cox 1982) (Fig. 12). Flow ZBC1 was dated by Stipp & McDougall (1968) at 8.24 ± 0.2 Ma: it is likely that the Church Formation was erupted during the normal magnetochron 4A (7.81–8.18 Ma). From the 7.03 ± 0.2 Ma age of the Kaioruru flow (ZBK1 or ZBK3) dated by Stipp & McDougall (1968), the Kaioruru flows can be correlated with the reversed subchron 4-1 (6.86 – 6.94 Ma, Cox 1982). Flow ZBK2, which has an intermediate direction, is near the bottom of the Kaioruru sequence and may have erupted at the beginning of the period of reversed polarity. The K-Ar dates for the Stoddart Formation all suggest an age of c. 6 Ma (Stipp & McDougall 1968; Evans 1970). The semi-intermediate direction recorded in nearly all the Stoddart Formation (this study and Evans 1970) suggests that the eruptions all occurred over a very short time, of the order of a few thousand years, either at the beginning or the end of a period of reversed polarity. From the K-Ar dates it is most likely that this part of the Stoddart Formation was formed at the end of chron 3Ar (6.07 Ma).

MEAN VIRTUAL GEOMAGNETIC POLES

The mean VGPs from Dunedin and Diamond Harbour are far sided (Wilson 1970). The mean Akaroa VGP is near sided, and the Lyttelton VGP is slightly right handed (Table 5, Fig. 13).

If the VGPs are rotated back to their original positions, using the hotspot-relative rotations of Livermore et al. (1983) (a rotation of 1.063°/Ma about an Euler pole at 63°.6N, 353°.94E was used), then the Dunedin and Lyttelton poles lie close to the geocentric axial dipole VGP, the Akaroa pole is still near sided, and the Diamond Harbour Group pole is far sided. The use of an alternative model for the hotspot-relative plate movement of eastern New Zealand, such as that of Gordon & Jurdy (1986) (a rotation of 0.767°/Ma about an Euler pole of 69°N, 292°E), makes little difference to the far and near sidedness of the Diamond Harbour and Akaroa VGPs.

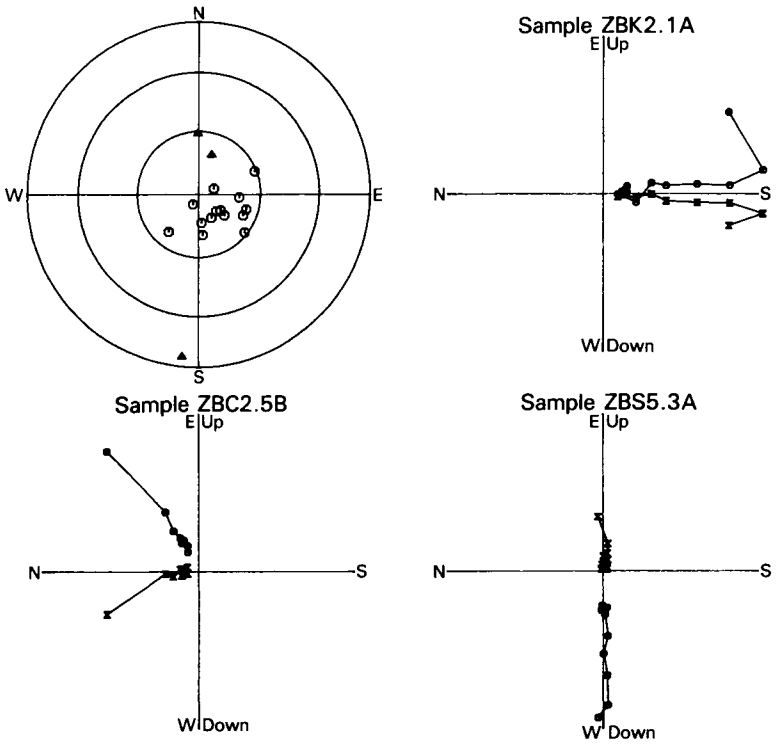


Fig. 11 Stereoplot showing the mean site NRMs for the three Diamond Harbour formations, along with examples of the demagnetisation data (0–50 mT in 5 mT steps. Symbols as in Fig. 2.

Table 4 Paleomagnetic and sampling data for the Diamond Harbour Group.

Site code	Grid ref.	Mean declination	Mean inclination	Polarity	n/N	α_{95}	k
ZBC1	099433	18.7	–70.3	N	3/6	7.5	270
ZBC2	074435	359.4	–61.0	N	3/5	5.4	526
ZBK1	078438	172.5	76.7	R	3/8	7.3	286
ZBK2	073435	186.1	–6.6	I	3/3	9.2	179
ZBK3	078438	173.2	70.9	R	3/4	10.1	149
ZBS1	097438	67.7	60.6	I	3/5	13.4	85
ZBS2	092440	124.1	77.0	R	3/3	12.1	105
ZBS3	079435	106.5	65.9	I	4/7	21.3	20
ZBS4	073437	69.1	82.1	I	4/5	13.3	49
ZBS5	070437	127.6	74.2	R	3/3	19.0	43
ZBS6	085442	128.5	61.3	R	3/5	28.4	20
ZBS7	078433	209.4	84.4	R	3/6	9.4	166
ZBS8	078433	131.9	78.5	R	3/5	24.3	27
ZBS9	078433	114.1	66.3	I	3/4	21.6	34
ZBS10	078433	93.8	70.3	I	3/5	13.5	85
ZBS11	078439	149.3	77.4	R	3/5	5.3	535
ZBS12	079436	219.5	67.5	R	3/4	21.3	34

Grid references refer to NZMS 1 series topographic map sheet S84. ZBC sites are Church, ZBK Kaioruru, and ZBS Stoddart.

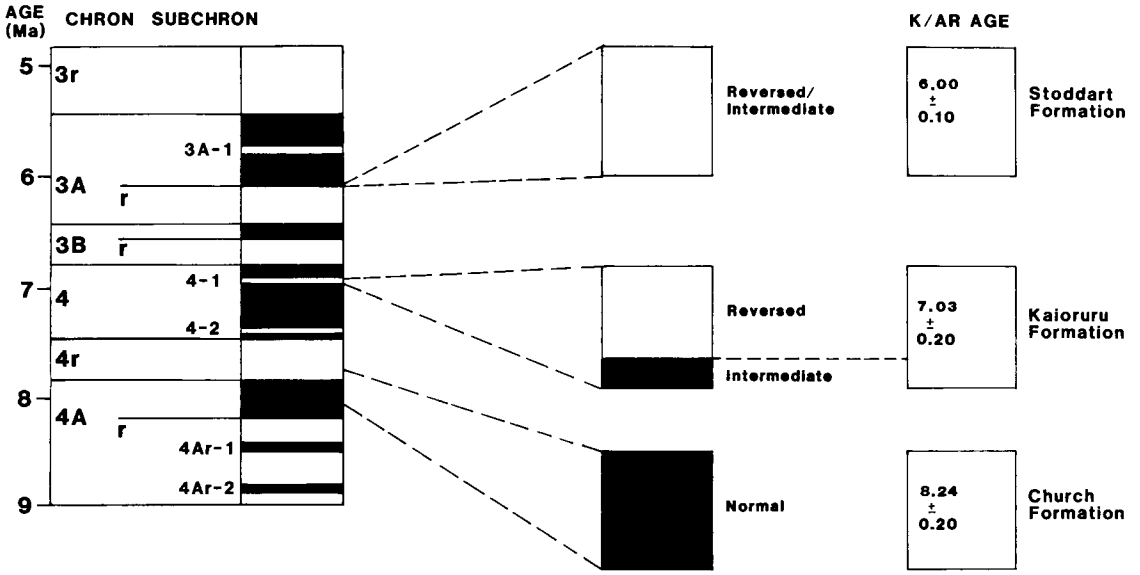


Fig. 12 A possible correlation of the Diamond Harbour Group magnetostratigraphy with the magnetic timescale (Cox 1982). K-Ar ages are recalculated from Stipp & McDougall (1968) and Evans (1970).

Table 5 Mean VGPs calculated from the New Zealand directional data: VGPs from reversed data have been normalised and intermediate directions excluded (except where stated).

Site		Latitude	Longitude	n	α_{95}	k	R
Dunedin	Normal	88.7N	035.3E	10	13.7	13.4	9.327
(45.°8S,	Reversed	80.2N	347.8E	4	24.6	15.0	3.799
170°.6E)	All	86.6N	357.0E	14	10.9	14.3	13.092
Lyttelton	Normal	80.0N	295.8E	5	15.6	25.1	4.841
(43°.6S,	Reversed	84.9N	099.9E	21	7.6	18.3	9.907
172°.75E)	All	87.7N	086.3E	26	6.9	18.0	24.642
Akaroa	Normal	84.6N	206.1E	12	10.2	18.9	11.419
(43°.8S,	Reversed	77.1N	130.8E	40	5.2	20.1	38.064
173°E)	All	79.7N	137.6E	52	4.7	18.7	49.269
	Evans (1970)	81.3N	158.5E	70	4.7	13.8	65.012
Diamond	Church	82.6N	308.2E	2	39.7	41.7	1.976
Harbour	Kaioruru	73.1N	005.1E	2	20.6	149.3	1.993
Group	Stoddart	61.7N	017.6E	7	16.5	14.3	6.581
(43°.6S,	All	68.6N	012.1E	11	11.7	16.1	10.379
172°.8E)	Stod. inc. int.	51.1N	028.8E	12	13.4	11.4	11.038
	Stod. (Evans)	61.1N	009.5E	44	2.3	90.1	43.523

Whereas the time-averaged geomagnetic field approximates to a geocentric axial dipole, Wilson & McElhinny (1974) have shown there is a tendency for late Tertiary paleomagnetic poles to be far sided. This is confirmed by recent studies (e.g., Livermore et al. 1983) which have shown that the best fitting model for the late Tertiary geomagnetic field includes quadrupole and octupole terms as well. However, the mean VGPs from these igneous centres show no consistent deviation towards near or far sidedness, so long-term nondipole fields are unlikely to be causing the observed deviations.

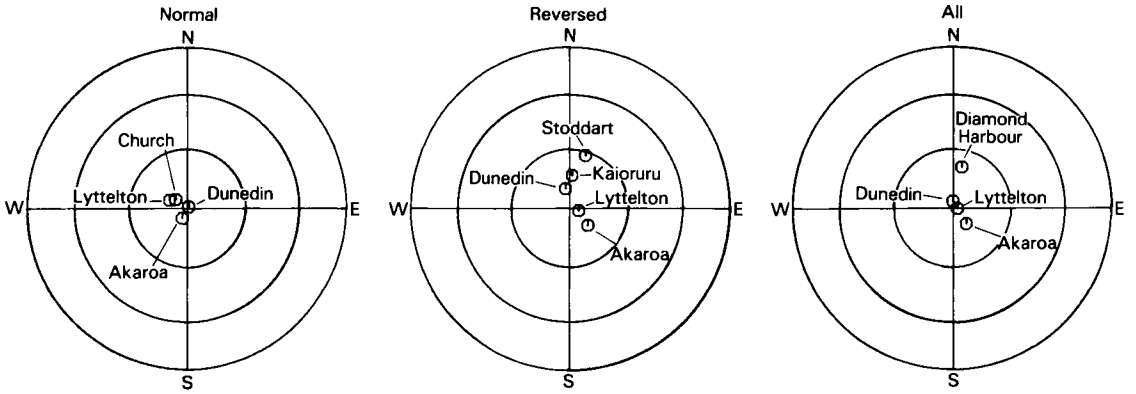


Fig. 13 The formation mean virtual geomagnetic poles (VGPs) (see Table 5). Reverse poles have been normalised and intermediate VGPs (which lie more than 40° from the poles) have been excluded. The reversed poles seem to deviate more from the axial dipole position.

Although all of the large volcanoes have probably undergone some subsidence since activity ceased (Wellman 1983; Weaver & Sewell 1986), there is no evidence of any major tectonic disturbance to the lava flows, and quaquaversal dips are preserved in all except for the western edge of the Dunedin volcano (Coombs 1983). Thus a tectonic origin for the far and near sidedness of the two younger centres may be discounted.

Although the fact that the reversed VGPs seem to be further from the geocentric axial dipole position might suggest that some of the rocks possess a hard secondary component that has not been isolated by AF demagnetisation up to only 50 mT, sister samples which were demagnetised up to 200 mT for paleointensity studies (Sherwood & Shaw 1986) showed no such evidence of a hard secondary component.

It appears therefore that the reason for the significant differences between the Diamond Harbour and Akaroa VGPs and the geocentric axial dipole VGP is that the interval sampled in each case was not long enough to cancel out the effects of secular variation. This is not surprising in the Diamond Harbour Group pole, as this is dominated by the semi-intermediate directions from the Stoddart Formation, which must have erupted in a short period of time. In the Akaroa sequence, several flows have recorded the same semitransitional direction immediately prior to the R-N reversal, suggesting that the interval between eruptions for these flows was only a few years, whereas the mean interval between flows for the Akaroa sequence must have been about 7000 years. Therefore the VGP mean is biased because of the irregularity of the igneous activity.

The mean VGPs from Dunedin and Lyttelton show that there has been no significant polar wander for the eastern part of New Zealand for the last 12 m.y., contrary to suggestions otherwise deduced from the Akaroa pole (Evans 1970). Despite the long time span of the Akaroa section (c. 0.5 m.y.), cancellation of secular variation does not seem to have occurred: this serves as a caution for anyone using poles from a series of lavas to form a polar wander path.

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